Pavel Rumiantsev

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Research interests: AI/ML, Neural Ahcitecture Search, Mixtures of Experts, Large Language Models, Agentic Systems

SUMMARY

AI/ML Researcher with 10+ years of experience in LLMs, generative AI, graph neural networks, and neural architecture search. Focused on experimental design, model evaluation, and scalable distributed experiments in multi-GPU environments. Published across NeurIPS, TMLR, and ICASSP, with work spanning LLM benchmarking, agentic systems, and graph neural networks. Experienced in building Proof of Concept and translating research into peer-reviewed reports.

SKILLS

Research: Experiment design, Applied research, Testing, Technical writing
AI/ML Research: experiment design, applied research, model evaluation, LLM benchmarking, statistical analysis, technical writing, deep learning, Transformer, MoE, GNNs, NAS
GenAI & LLM: RAG, Multi-Agent Systems, Speculative decoding, OpenAI API, MCP Tools, Claude Code
ML Engineering: Python, Golang, SQL, Docker, AWS, GCP, CI/CD, FastAPI, Ray, Slurm, Hugging Face, vLLM
Frameworks: PyTorch, Pandas, scikit-learn, NumPy, SciPy, Deep Graph Library (DGL)
Leadership: Mentoring, Cross-Functional Collaboration, Research Communicationg
Languages: English (Advanced), Russian (Native), French (Beginner)

WORK EXPERIENCE

AI Research Scientist

Huawei Noah's Arc Lab, Montreal, QC, Canada — *Jun 2023 - Feb 2026*

- Led research on AI, GNN, LLMs, and Agentic systems, resulting in **4 peer-reviewed publications** (including 2 **NeurIPS papers**)
- Owned research direction **Graph Neural Networks** and **Mixture-of-Experts** architectures, publishing novel methods for graph knowledge distillation and MoE routing.
- Built and maintained research prototypes in Python using **OpenAI API and vLLM**, supporting efficient inference, reasoning, and agentic workflows.
- Architected a distributed experimentation framework using **Ray and FastAPI**, orchestrating workloads on **8-GPUs clusters**, capable of running **500+ inference experiments**.
- Owned design and delivery cost-aware multi-LLM routing and inference pipelines, **benchmarking 15+ proprietary and open-source LLMs** across **12 reasoning, mathematics, and comprehension datasets**.
- Authored literature reviews, technical reports, and research papers, synthesizing findings from **200+ research publications** across AI, ML, and LLM domains.
- Collaborated with **5+ engineering teams** to transform research prototypes into production-ready AI solutions.
- **Mentored 2 junior researchers** sharing best practices in AI research and solution design

AI Researcher

6th Grain, Montreal, QC, Canada — *Sep 2021 - Nov 2022*

- Owned **end-to-end development AI and ML solutions** for precision agriculture using satellite imagery and deep learning.
- Designed and trained computer vision models **GCP Vertex AI** for crop classification and segmentation using Python and PyTorch, **improving model accuracy by 46%** compared to baseline approaches.
- Directed processing and analysis of **14+ TB of geospatial data** using scalable preprocessing pipelines and leveraging **GCP Cloud Storage** for data management and **GCP Maps API** for geospatial referencing and visualization.
- Built and **owned reproducible ML training pipelines** including data aggregation, preprocessing, feature engineering, model training, and evaluation, **reducing update time by 45%** through workflow automation.
- Authored technical analyses and benchmarking reports comparing model performance across **10+ datasets**, guiding model selection and deployment decisions.
- Established machine learning best practices including experiment tracking, data governance, and **reproducible model evaluation** to improve reliability and maintainability.

Graduate AI Researcher

McGill Networks Research Lab, Montreal, QC, Canada — *Sep 2019 – Aug 2025*

- Led research on Neural Architecture Search, Mixture-of-Experts, and efficient LLM inference, resulting in **6 peer-reviewed publications (including 2 NeurIPS papers)**.
- Owned design and implementation of a novel **Mixture-of-Experts** architecture for GNN-to-MLP knowledge distillation in **PyTorch and Deep Graph Library**, ranking **first across 8 graph datasets**.
- Designed and deployed a **Retrieval-Augmented Generation (RAG)** powered AI research assistant, delivering grounded, citation-backed responses over internal research documentation.
- Architected a zero-shot/one-shot NAS search pipeline and statistical ranking-evaluation framework in Python using DARTS and NAS-Bench APIs, **cutting GPU memory use by 81%**.
- Built reproducible experiment infrastructure with version-controlled configs, automated tests, and experiment tracking, **supporting 10+ collaborative codebases.**
- Led transfer of sparse GNN decomposition research using Deep Graph Library, **cutting inference latency by 88%** in production-bound graph models.
- **Mentored junior lab members** and reviewed conference submissions sharing practices for ML research, including testing, packaging, and reproducibility.

Machine Learning Engineer

PolySoft, Moscow, Russia *Oct 2017 - July 2019*

- Built and owned machine learning models in Python and PyTorch for UAV recognition and real-time tracking, achieving inference latency less than **50 ms** while maintaining **99%** detection rate and **97%** recognition accuracy.
- Architected computer vision pipelines for deployment on resource-constrained hardware (**Nvidia Jetson TX2**).
- Built and maintained internal automated data preprocessing and annotation pipelines processing **80,000+ images and video frames**, accelerating model iteration and **reducing manual labelling effort by 85%**.
- Led evaluation and benchmarking **10+ machine learning architectures**, improving robustness across varying operating conditions.
- Drove solution design collaboration with **product and hardware teams** on solution design, integrating ML components into larger enterprise embedded systems.
- Owned CI/CD pipeline upgrades, improving **deployment productivity by 30%**

EDUCATION

McGill University **Montreal, QC, Canada**

PhD in Electrical Engineering *Sep 2019 - Feb 2026*

- Thesis title: "*Efficient Algorithms for Automatic Structure Identification and Adaptation in Deep Neural Networks*"

National Research University Higher School of Economics **Moscow, Russia**

Master's degree in Applied Mathematics *Sep 2017 - May 2019*

- Thesis title: "*Research and Development of the Neural System Components for Flying Objects Recognition*"

Moscow Technological University **Moscow, Russia**

Bachelor's degree in Control Systems Engineering *Sep 2013 - May 2017*

- Thesis title: "*Fuzzy Controller with Self-Adjusting Membership Functions for Servomotor Position Control*"

PRESENTATIONS & TALKS

- Poster session at NeurIPS 2025 LLM Evaluation Workshop *2025*
- Presentation at Graph Signal Processing Workshop on Graph Knowledge Distillation to Mixture of Experts *2025*
- Presentation at ILLS research seminar on Neural Architecture Search *2024*
- Presentation at ICASSP 2023 on efficient Zero-Shot Neural Architecture Search *2023*

PUBLICATIONS

- Pavel Rumiantsev, Soumyasundar Pal, Yingxue Zhang, and Mark Coates. FEval-TTC: Fair Evaluation Protocol for Test-Time Compute. In *Proc. NeurIPS 2025 Workshop Eval. Evol. LLM Lifecycle* (NeurIPS), Dec. 2025 Link
- Antonios Valkanas, Soumyasundar Pal, Pavel Rumiantsev, Yingxue Zhang, and Mark Coates. C3PO: Optimized Large Language Model Cascades with Probabilistic Cost Constraints for Reasoning. In *Adv. Neural Inf. Process. Syst.* (NeurIPS), Dec. 2025 Link
- Pavel Rumiantsev and Mark Coates. Variation matters: from mitigating to embracing Zero-Shot NAS ranking function variation. *Trans. Mach. Learn. Res.* (TMLR), Mar. 2025 Link
- Yaochen Hu, Mai Zeng, Ge Zhang, Pavel Rumiantsev, Liheng Ma, Yingxue Zhang, and Mark Coates. Sparse Decomposition of Graph Neural Networks. *Trans. Mach. Learn. Res.* (TMLR), Mar. 2025 Link
- Pavel Rumiantsev and Mark Coates. Graph Knowledge Distillation to Mixture of Experts. *Trans. Mach. Learn. Res.* (TMLR), Oct. 2024 Link
- Pavel Rumiantsev and Mark Coates. Performing Neural Architecture Search without gradients. In *Proc. IEEE Int. Conf. Acoust. Speech, Signal Proc.* (ICASSP), Jun. 2023 Link

PROJECTS

- **FEval-TTC**, Fair Evaluation Protocol for Test-Time Compute. This LLM evaluation framework on Python features carefully engineered Chain-of-Thought few-shot prompting protocols for multiple LLMs on a variety of mathematical and reasoning datasets. The few-shot query process and answer extraction are standardized across the datasets, which eases the burden on researchers in terms of time and money. GitHub
- **C3PO-LLM**, Cost Controlled Cascaded Prediction Optimization for Large Language Models. This self-supervised multi-LLM Python framework optimizes inference under cost constraints. It is compatible with any prompt engineering techniques for LLM predictions. This repository includes the official implementation as well as the cascade baselines. GitHub
- **Routing-by-Memory**, official code for "Graph Knowledge Distillation to Mixture of Experts". A router design enforces expert specialization by making each expert to specialize on a certain region on the hidden representation space. The project is made on Python. GitHub
- **ETENAS**, the fast zero-shot neural architecture search ranker that does not use gradients. The project is made on Python. GitHub
- **EvolveGCN**, Python implementation of the paper "Evolving Graph Convolutional Networks for Dynamic Graphs" applying it to Elliptic Bitcoin dataset. GitHub
- **GraphNetLib**, the graph networks processing Python library for Tensorflow 2.x. GitHub
- **Toy blockchain**, example of blockchain on Golang for concept demonstration purposes. The project was made to demonstrate a set of principles behind Bitcoin and it follows bitcoin protocols. GitHub